To find minima using bisection method

$$f(x) = \frac{a}{x} + x^b$$

$a \rightarrow$ 50 to 54 (int)

$b \rightarrow$ 2 to 3 (int)

Take derivative of $f(x)$.

$$f'(x) = -\frac{a}{x^2} + b x^{b-1}$$

Take initial guess $x_1 = 2$ & $x_2 = 5$

$f'(x_1) = \boxed{}$ $-$ $\qquad$ $f'(x_2) = \boxed{}$ $+$

Iteration 1

$$x_3 = \frac{x_1 + x_2}{2} = \boxed{3.5}$$

$f'(x_3) = \boxed{}$ $+$ve

If $f'(x_1) f'(x_3) < 0$ since $f'(x_1) \times f'(x_3) < 0$ Root lies betn x_1 & x_3

else since $f'(x_2) \times f'(x_3) < 0$ Root lies " x_2 & x_3

If $f'(x_1) \times f'(x_3) < 0$

$x_1 = \boxed{} \Rightarrow x_1 = 2$

$x_2 = \boxed{} \Rightarrow x_3 = 3.5$

else

$x_1 = \boxed{} \Rightarrow x_3 = 3.5$

$x_2 = \boxed{} \Rightarrow x_2 = 5$

Iteration 2

$$x_3 = \frac{x_1 + x_2}{2} = \boxed{}$$

Repeating this procedure

$f'(x_1) \times f(x_3) < 0$

x_1	$f'(x_1)$	x_2	$f'(x_2)$	$x_3 = (x_1 + x_2)/2$	$f'(x_3)$
2	✓	5	✓	✓	✓
2	✓	✓	✓	✓	✓

$\downarrow$

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